MIL-F-52525/5A
30 June 1987
SUPERSEDING
MIL-F-52525/5
11 October 1978

MILITARY SPECIFICATION SHEET

FITTINGS, WIRE REINFORCED HYDRAULIC HOSE AND CLAMP-HALVES: FIELD ATTACHABLE (REUSABLE), SCREW-ON TO 37° FLARE FEMALE SWIVEL, STRAIGHT, TYPE 100R1, FOR USE WITH TYPE 100R1 SINGLE WIRE BRAID HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for requiring the fittings described herein shall consist of this specification and the latest issue of MIL-F-52525.

REQUIREMENTS:

1. The fitting shall conform to figure 1 and table I.

AMSC N/A
FSC 4730
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TABLE I. Dimensions for type 100R1 fittings.1/

Specification part number	Hose dash number	Nominal hose I.D.	Tube dash number	Nominal tube O.D.	Nominal thread size	H Nominal hex size	L Maximum	
		inch		inch		inch	inch	mm
M52525/5-4-4	4	1/4	4	1/4	7/16-20	9/16	0.87	22.09
M52525/5-4-5	4	1/4	5	5/16	1/2-20	5/8	0.87	22.09
M52525/5-6-6	6	3/8	6	3/8	9/16-18	11/16	1.09	27.68
M52525/5-8-8	8	1/2	8	1/2	3/4-16	7/8	1.20	30.48
M52525/5-8-10	8	1/2	10	5/8	7/8-14	1	1.20	30.48
M52525/5-12-12	12	3/4	12	3/4	1-1/16-12	1-1/4	1.59	40.38
M52525/5-16-16	16	1	16	1	1-5/16-12	1-1/2	2.02	51.30
M52525/5-20-20	20	1-1/4	20	1-1/4	1-5/8-12	2	2.46	62.48

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TABLE I. Dimensions for type 100R1 fitting.1/ (Continued)

Specification part number	Hose dash number	Nominal hose I.D.	Tube dash number	Nominal tube O.D.	Nominal thread size	M Mimimum		N	
		inch		inch		inch	mm	inch +0.030 -0.150	mm +.76 -0.40
M52525/5-4-4	4	1/4	4	1/4	7/16-20	0.300	7.62	0.344	8.74
M52525/5-4-5	4	1/4	5	5/16	1/2-20	0.331	8.41	0.375	9.52
M52525/5-6-6	6	3/8	6	3/8	9/16-18	0.333	8.46	0.375	9.52
M52525/5-8-8	8	1/2	8	1/2	3/4-16	0.375	9.52	0.422	10.72
M52525/5-8-10	8	1/2	10	5/8	7/8-14	0.461	11.70	0.500	12.70
M52525/5-12-12	12	3/4	12	3/4	1-1/16-12	0.469	11.91	0.562	14.27
M52525/5-16-16	16	1	16	1	1-5/16-12	0.563	14.30	0.594	15.09
M52525/5-20-20	20	1-1/4	20	1-1/4	1-5/8-12	0.563	14.30	0.625	15.87

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TABLE I. Dimensions for type 100R1 fitting.1/ (Continued)

Specification part number	Hose dash number	Nominal hose I.D. inch	Tube dash number	Nominal tube O.D. inch	Nominal thread size	P inch ±0.005	P mm ±0.13	Q Minimum inch	Q Minimum mm
M52525/5-4-4	4	1/4	4	1/4	7/16-20	0.295	7.49	0.110	2.790
M52525/5-4-5	4	1/4	5	5/16	1/2-20	0.355	9.02	0.110	2.790
M52525/5-6-6	6	3/8	6	3/8	9/16-18	0.435	11.04	0.260	6.600
M52525/5-8-8	8	1/2	8	1/2	3/4-16	0.570	14.48	0.375	9.650
M52525/5-8-10	8	1/2	10	5/8	7/8-14	0.680	17.27	0.375	9.650
M52525/5-12-12	12	3/4	12	3/4	1-1/16-12	0.850	21.59	0.570	14.480
M52525/5-16-16	16	1	16	1	1-5/16-12	1.100	27.94	0.780	19.810
M52525/5-20-20	20	1-1/4	20	1-1/4	1-5/8-12	1.410	35.81	1.010	25.650

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TABLE I. Dimensions for type 100R1 fitting.1/ (Continued)

Specification part number	Hose dash number	Nominal hose I.D. inch	Tube dash number	Nominal tube O.D. inch	Nominal thread size	C Maximum inch	C Maximum mm
M52525/5-4-4	4	1/4	4	1/4	7/16-20	2.62	66.55
M52525/5-4-5	4	1/4	5	5/16	1/2-20	2.62	66.55
M52525/5-6-6	6	3/8	6	3/8	9/16-18	2.78	70.61
M52525/5-8-8	8	1/2	8	1/2	3/4-16	3.24	82.30
M52525/5-8-10	8	1/2	10	5/8	7/8-14	3.34	83.06
M52525/5-12-12	12	3/4	12	3/4	1-1/16-12	3.68	93.47
M52525/5-16-16	16	1	16	1	1-5/16-12	4.28	108.71
M52525/5-20-20	20	1-1/4	20	1-1/4	1-5/8-12	4.96	125.98

1/ All dimensions not shown shall conform to SAE J516.

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2. The maximum operating pressures shall be as specified in table II.

TABLE II. Maximum operating pressures of fittings.

Hose I.D. (REF) (inch)	PSI (pounds per square inch)	MPa (megapascals)
1/4	2750	19.0
5/16	2500	17.2
3/8	2250	15.5
1/2	2000	13.8
5/8	1500	10.3
3/4	1250	8.6
1	1000	6.9
1-1/4	625	4.3

3. The over-tightening torque for swivel nuts shall be as specified in table III.

TABLE III. Over-tightening torque for swivel nut.

Nominal tube O.D. (REF) (inch)	Torque	
	inch pounds	Newton meter (Nm)
1/4	135	15.3
5/16	180	20.3
3/8	270	30.5
1/2	450	50.8
5/8	650	73.4
3/4	900	101.7
1	1200	135.6
1-1/4	1900	214.7

1/ Torque is expressed as Newton meter per radian $(N \cdot m/rad) \cdot rad = N \cdot m$.

4. Marking shall include the manufacturer's name of trademark, size, and hose identifier R-1. Location of marking is optional.

5. Method of fabrication and assembly of swivel nut is optional.

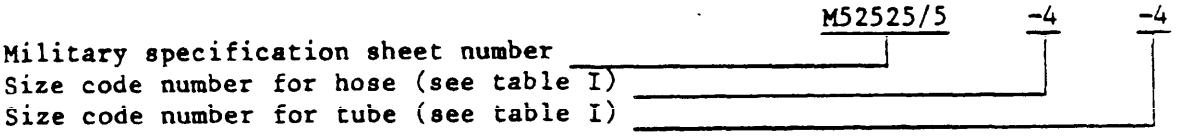
6. This item must be listed on QPL-52525.

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INTENDED USE: Type 100R1 fittings are for use with single wire braid reinforced hose conforming to MIL-H-52471, type 100R1.

CATALOGING DATA: The military specification part number is a definitive part number which corresponds to the size fitting covered by this specification sheet. The military specification number, the military specification sheet number, code number for size of hose, and code number for size of tube are combined to form the military specification part number.

EXAMPLE: Military specification part number



The approved item name of this item, in accordance with the Federal Item Name Directory for the Federal Catalog System, Handbook H6, Section A, is Adapter, Tube to Hose.

Custodian:
Army - ME

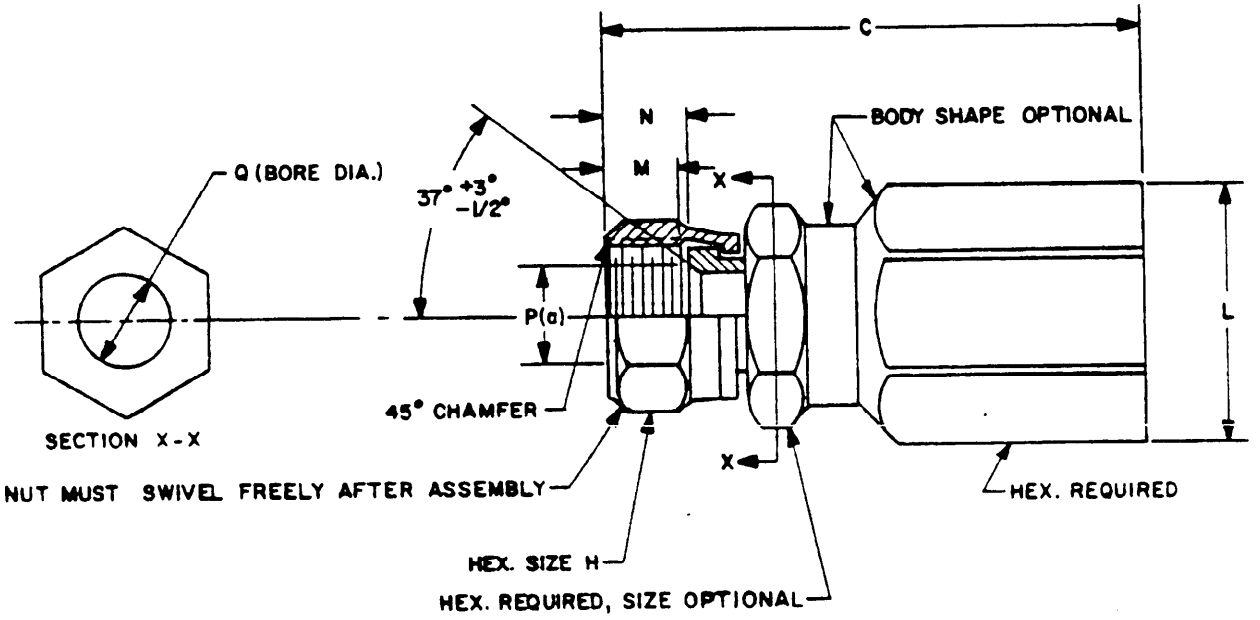
Review activities:
Army - GL
DLA - CS

User activity:
Air Force - 99

Preparing activity:
Army - ME

Project 4730-0521

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(a) DIMENSION P TO BE CONCENTRIC WITH DIMENSION Q WITHIN 0.015 F I M.

FIGURE 1. Fitting, screw-on to 37° flare, straight.

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